FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This Final Explanation document is your model answer for the driving question. Read it carefully, then use it as a guide to write your own full explanation in your exercise book or on the platform. Your goal is to show that you truly understand what happens to lengths, areas, and volumes when a shape or solid is scaled — and that you can use this understanding to solve real problems like those faced by builders, engineers, and planners right here in Kenya.

DRIVING QUESTION: When you scale up a shape or solid — like doubling the size of a water tank or enlarging a building plan — what exactly happens to its lengths, its area, and its volume, and how can we use this to solve real problems in our community?

HOW TO USE THIS DOCUMENT:

1. Read each section carefully and notice how every claim is supported with mathematics.
2. Pay attention to how the Kisumu school architectural plan and the two water tanks are used as evidence throughout — you should reference the anchoring phenomenon in your own answer too.
3. Use the worked examples as models, but write your own calculations using different numbers.
4. Check your finished explanation against the rubric at the bottom of this document before you submit.

Section 1 — What Is a Scale Factor and Why Does It Matter?

Explain what a scale factor is, where we see it in real life in Kenya, and how to calculate the real measurement from a scale drawing.	A scale factor is the single number that tells you how many times larger (or smaller) a real object is compared to its drawing or model. When the architectural plan in the Kisumu school staffroom is labelled '1 : 200,' it means every 1 centimetre on the paper represents 200 centimetres (2 metres) in real life. The scale factor for lengths from the drawing to the real building is therefore 200. You will meet scale factors in many Kenyan contexts: a road map of Nairobi might use a scale of 1 : 50 000, a carpenter in Kericho sketches a door frame at 1 : 10, and a jua-kali artisan in Kisumu uses a scaled template to cut metal sheets for water tanks. To move from drawing measurement to real measurement, multiply by the scale factor (k):

	Real length = Drawing length × k Worked Example: The architectural plan shows the school's main classroom block as 5 cm wide. The scale is 1 : 200. Real width = 5 cm × 200 = 1 000 cm = 10 metres. ✓ To move from real measurement to drawing measurement, divide by k: Drawing length = Real length ÷ k Key vocabulary you must use in your answer: scale factor (k), ratio, proportion, similar figures.

Section 2 — Similar Shapes: Lengths Scale by k, But Angles Stay the Same	
Explain the conditions for two shapes to be mathematically similar, and describe what happens to corresponding lengths when a shape is enlarged by scale factor k.	Two shapes are mathematically similar if (i) all corresponding angles are equal, and (ii) all corresponding lengths are in the same ratio. That ratio is the scale factor k. Look at the school plan again. Every room on the drawing has exactly the same angles as the real room — all the right-angle corners of a classroom are still 90° on paper. What changes is size, not shape. This is the heart of similarity. If shape A is enlarged by scale factor k to produce shape B, then: Every length in B = k × corresponding length in A. Worked Example: On the Kisumu school plan, the football pitch measures 5 cm × 2.5 cm. Scale is 1 : 200. Real length = 5 × 200 = 1 000 cm = 10 m Real width = 2.5 × 200 = 500 cm = 5 m Both dimensions were multiplied by the same k = 200, so the pitch on the plan is similar to the real pitch. Important: Perimeter also scales by k. If the drawing perimeter is 15 cm, the real perimeter is 15 × 200 = 3 000 cm = 30 m. Remember: doubling every length (k = 2) does NOT double the area — that is the misconception the water tanks will help you correct in the next section.

Section 3 — Area Scales by k^2, Not by k	
Explain and prove why area scales by the square of the scale factor, and apply this to a real context such as floor area	Here is where many students make an error. If you double every length of a shape, your instinct might say the area doubles too. The water tanks at the Kisumu school show us this is WRONG.

or a farm plot.	Proof using a simple rectangle: Suppose a rectangle has length L and width W. Original area = L × W Now scale every length by k: New length = kL, New width = kW New area = kL × kW = k^2LW = k^2 × (original area) So area scales by k^2 (k squared), not k. Worked Example — School Classroom Floor: The plan shows one classroom as 3 cm × 2 cm. Scale k = 200. Real dimensions: 6 m × 4 m Real floor area = 6 × 4 = 24 m^2 Using the area scale factor directly: Drawing area = 3 × 2 = 6 cm^2 Area scale factor = k^2 = 200^2 = 40 000 Real area = 6 cm^2 × 40 000 = 240 000 cm^2 = 24 m^2 ✓ Real-Life Application — Farm Plot in Kericho: A tea farmer in Kericho has a square plot of side 50 m (area = 2 500 m^2). She wants to plant a second plot that is similar but with side 100 m (k = 2). New area = 2^2 × 2 500 = 4 × 2 500 = 10 000 m^2 The area is 4 times larger, not 2 times. She will need 4 times as many tea seedlings, not 2 times. Key rule to memorise: Area Scale Factor = k^2

Section 4 — Volume Scales by k^3, Not by k or k^2	
Explain and prove why volume scales by the cube of the scale factor, and use the two Kisumu school water tanks as your central evidence.	The two water tanks at the Kisumu school are the perfect evidence for this idea. The taller tank appears twice as tall as the shorter tank — meaning k = 2 for lengths. Yet it holds 8 times as much water. Why? Proof using a rectangular box (cuboid): Suppose a tank has length L, width W, and height H. Original volume = L × W × H Scale every dimension by k: New volume = kL × kW × kH = k^3 × L × W × H = k^3 × original volume So volume scales by k^3 (k cubed).

	For the school tanks, $k = 2$: Volume scale factor = $2^3 = 8$ The larger tank holds 8 times the volume. ✓ — exactly what the students observed! This resolves the cognitive conflict from the start of our investigation. Doubling every length does NOT double the volume; it multiplies the volume by 8. Worked Example — Cylindrical Water Tank: A small cylindrical tank in a home compound in Mombasa has radius 0.5 m and height 1 m. Volume = $\pi \times (0.5)^2 \times 1 \approx 0.785$ m^3 A hardware shop sells a 'double-size' tank ($k = 2$): radius 1 m, height 2 m. Volume = $\pi \times (1)^2 \times 2 \approx 6.283$ m^3 Ratio = $6.283 \div 0.785 = 8 = 2^3$ ✓ Real-World Implication: If a community in the Rift Valley wants a tank that holds 8 times more water, they only need to double the linear dimensions — not build a tank 8 times taller. This is crucial knowledge for engineers, jua-kali fabricators, and community water projects. Key rule to memorise: Volume Scale Factor = k^3

Section 5 — Putting It All Together: Answering the Driving Question	
Write a complete, connected answer to the driving question that links scale factor, area, and volume using evidence from the Kisumu school phenomenon, and explain at least two ways this mathematics is useful in the Kenyan community.	FULL MODEL ANSWER TO THE DRIVING QUESTION: When you scale up a shape or solid by a linear scale factor k, three different things happen depending on what you are measuring: 1. LENGTHS scale by k. Every distance — width, height, perimeter, diagonal — is multiplied by k. On the Kisumu school plan ($k = 200$), a 5 cm line on the drawing represents a 10-metre wall in real life. This is why every classroom, corridor, and window appears in correct proportion: the same scale factor applies to every length equally. 2. AREAS scale by k^2. Because area involves two length dimensions multiplied together, the scale factor is applied twice: $k \times k = k^2$. If you double every length of a classroom floor ($k = 2$), the floor area becomes $2^2 = 4$ times larger. A farmer in Kericho doubling the side of a plot gets 4 times the planting area and needs 4 times the seeds and fertiliser — not 2 times. 3. VOLUMES scale by k^3. Because volume involves three length dimensions, the scale factor is applied three times: $k \times k \times k = k^3$. This is exactly why the

	taller water tank at the Kisumu school ($k = 2$) holds $2^3 = 8$ times more water than the shorter tank, even though it looks 'only twice as big.' Doubling every dimension multiplies volume by 8, not 2. SUMMARY TABLE: Measurement | Scale Factor Length | k Area | k^2 Volume | k^3 COMMUNITY APPLICATIONS: Application 1 — Reading Architectural Plans: Architects and building committees across Kenya use scale drawings to plan schools, hospitals, and housing. Knowing the linear scale factor (k) allows community members to calculate real lengths, floor areas (k^2) for tiling or roofing materials, and even tank volumes (k^3) for water storage planning — all from a single A3 drawing. Application 2 — Water Tank Sizing for Rural Communities: Many community groups in the Rift Valley and Western Kenya need to know: 'If we buy a tank with double the height, how much more water does it hold?' Without understanding k^3, they might underestimate and run out of water, or overspend on a tank that is far larger than needed. The scale factor for volume (k^3) gives an exact, reliable answer. Application 3 — Jua-Kali Metal Fabrication: Artisans in Kisumu and Nairobi who fabricate water tanks, cooking pots, or roofing sheets use similarity and scale factors daily. A client who wants a sufuria (cooking pot) that holds 8 times more ugali does not need a pot 8 times taller — they need one with dimensions scaled by $k = 2$ (since $2^3 = 8$). Understanding this saves material, time, and money. CONCLUSION: The Kisumu school plan and its two water tanks are not just classroom examples — they are everyday evidence of a powerful mathematical truth. Scaling is not a simple 'multiply everything by the same number' story. Lengths, areas, and volumes each respond to scale in their own way, and understanding the difference between k, k^2, and k^3 is an essential tool for anyone involved in building, farming, engineering, or community development in Kenya.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**

Understanding of Scale Factor (k) and Similarity	Accurately defines scale factor and similarity conditions (equal angles, proportional sides). Correctly calculates real and drawing lengths using k. Explicitly connects to the Kisumu architectural plan with correct working.	Correctly defines scale factor and uses it to convert between drawing and real measurements. Mentions the school plan but may not fully explain similarity conditions.	Shows a partial understanding of scale factor. May confuse the direction of scaling (drawing→real vs real→drawing) or omit the similarity conditions. Connection to the phenomenon is weak or absent.
Area Scale Factor (k^2) — Explanation and Application	Clearly proves and states that area scales by k^2, not k. Provides a correct worked example with accurate arithmetic. Applies the concept to a relevant Kenyan context (e.g. farm plot, classroom floor, roofing) and explains the practical implication.	States that area scales by k^2 and provides a mostly correct worked example. The Kenyan context is mentioned but the practical implication may not be fully developed.	Attempts to address area scaling but may state it scales by k (the common misconception) or apply k^2 inconsistently. Worked example contains errors or is missing.
Volume Scale Factor (k^3) — Explanation and Application	Clearly proves and states that volume scales by k^3. Uses the two Kisumu school water tanks as direct evidence ($k = 2$, volume ratio = 8). Provides an additional correct worked example. Explains why this matters for water storage or construction in the community.	States that volume scales by k^3 and references the water tanks. Provides a correct or mostly correct worked example. Practical implication is mentioned but may lack depth.	Attempts to address volume scaling but may confuse k^3 with k or k^2. The water tank evidence is not used or is incorrectly interpreted. Worked example is missing or contains significant errors.
Mathematical Reasoning, Accuracy, and Use of Proof	All calculations are correct and clearly shown step-by-step. Uses algebraic reasoning (e.g. $kL \times kW = k^2LW$) to prove — not just state — the scale factor rules. Correct mathematical vocabulary is used consistently throughout (scale factor, similar, ratio, proportion, area, volume).	Most calculations are correct and working is shown. At least one proof or derivation is attempted. Mathematical vocabulary is mostly used correctly with minor inconsistencies.	Some calculations are correct but working is incomplete or hard to follow. Rules are stated without proof or justification. Mathematical vocabulary is used sparingly or incorrectly.
Real-World Application and Connection to the Kenyan Community	Identifies and explains at least two distinct, specific applications of similarity and scale factors in a Kenyan context (e.g. architectural plans, water tank sizing, jua-kali fabrication, farming). For each, clearly explains HOW the mathematics (k, k^2, or k^3) is used and WHY it matters to people in the community.	Identifies at least two Kenyan applications and explains how scale factors are used in each. The explanation of community impact is present but may be general rather than specific.	Mentions one or two applications but the link to the mathematics is vague. The Kenyan community context is mentioned in passing without meaningful explanation of how the scale factor rules are applied.